

Differential Calculus — Chain Rule

Chapter 2: Power of Trigonometric Function | Page 33

PROBLEM STATEMENT

Trigonometric Power & Chain Rule

Find the derivative with respect to x of the function:

$$y = \sin^3 x$$

STEP 1 প্রদত্ত ত্রিকোণমিতিক ঘাত ফাংশনটি উল্লেখ করি (Given Function)

$$y = \sin^3 x = (\sin x)^3$$

STEP 2 উভয়পক্ষে x -এর সাপেক্ষে অন্তরীকরণ করি (Differentiating both sides w.r.t. x)

$$\frac{dy}{dx} = \frac{d}{dx} [(\sin x)^3]$$

STEP 3 শক্তি বিধি ও চেইন রুল প্রয়োগ করি ($\frac{d}{dx}[u^n] = nu^{n-1} \frac{du}{dx}$)

$$\frac{dy}{dx} = 3(\sin x)^{3-1} \cdot \frac{d}{dx}(\sin x) = 3 \sin^2 x \cdot \frac{d}{dx}(\sin x)$$

STEP 4 অভ্যন্তরীণ সাইন ফাংশনের অন্তরক নির্ণয় করি (Differentiating Base Function)

$$\frac{d}{dx}(\sin x) = \cos x$$

STEP 5 মান বসিয়ে চূড়ান্ত অন্তরজ পাই (Final Product Form)

$$\frac{dy}{dx} = 3 \sin^2 x \cos x$$

FINAL ANSWER
$$\frac{dy}{dx} = 3 \sin^2 x \cos x$$

Rigorous Insights & Mark Scheme

AMIE Section-A Standard Evaluation Protocol | Page 33, HW 06

🔴 BOX 1: FORMULATION & TRIGONOMETRIC POWER RULES

- **General Power Formula:** $\frac{d}{dx}[\sin^n x] = n \sin^{n-1} x \cos x$.
- **Alternative Double-Angle Form:**

$$3 \sin^2 x \cos x = \frac{3}{2} \sin x (2 \sin x \cos x) = \frac{3}{2} \sin x \sin 2x$$

- **Triple-Angle Identity Alternative:** Since $\sin 3x = 3 \sin x - 4 \sin^3 x \implies \sin^3 x = \frac{1}{4}(3 \sin x - \sin 3x)$.

$$\frac{dy}{dx} = \frac{1}{4}(3 \cos x - 3 \cos 3x) = \frac{3}{4}(\cos x - \cos 3x)$$

⚠️ BOX 2: COMMON STUDENT TRAPS & PITFALLS

- **Confusing $\sin^3 x$ with $\sin(x^3)$:** $\frac{d}{dx}(\sin^3 x) = 3 \sin^2 x \cos x$, whereas $\frac{d}{dx} \sin(x^3) = 3x^2 \cos(x^3)$. Always identify whether the exponent belongs to the function or the variable.
- **Stopping at $3 \sin^2 x$:** Omitting the inner derivative $\frac{d}{dx}(\sin x) = \cos x$.

🌐 BOX 3: REAL DOMAIN & EXTREME VALUES

- **Domain & Range:** Domain is $\mathbb{R}$; Range is $[-1, 1]$.
- **Critical Points:** $\frac{dy}{dx} = 0 \iff 3 \sin^2 x \cos x = 0$, which occurs when $\sin x = 0$ ($x = k\pi$) or $\cos x = 0$ ($x = \frac{\pi}{2} + k\pi$).
- At $x = k\pi$, the curve has stationary inflection points; at $x = \frac{\pi}{2} + k\pi$, local extrema occur.

💡 BOX 4: LEIBNIZ SUBSTITUTION (ALTERNATIVE VERIFICATION)

- Let $u = \sin x \implies y = u^3$.
- $\frac{dy}{du} = 3u^2$.
- $\frac{du}{dx} = \cos x$.
- By Chain Rule:

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = 3u^2 \cdot \cos x = 3 \sin^2 x \cos x$$

📊 BOX 5: AMIE EXAMINATION MARKS DISTRIBUTION (TOTAL: 3 / 4 MARKS)

- **1.0 Mark:** Correct setup writing $y = (\sin x)^3$ and applying power rule $3(\sin x)^2$.
- **1.0 Mark:** Applying inner derivative $\frac{d}{dx}(\sin x) = \cos x$.
- **1.0 Mark:** Clear algebraic packaging as $3 \sin^2 x \cos x$.